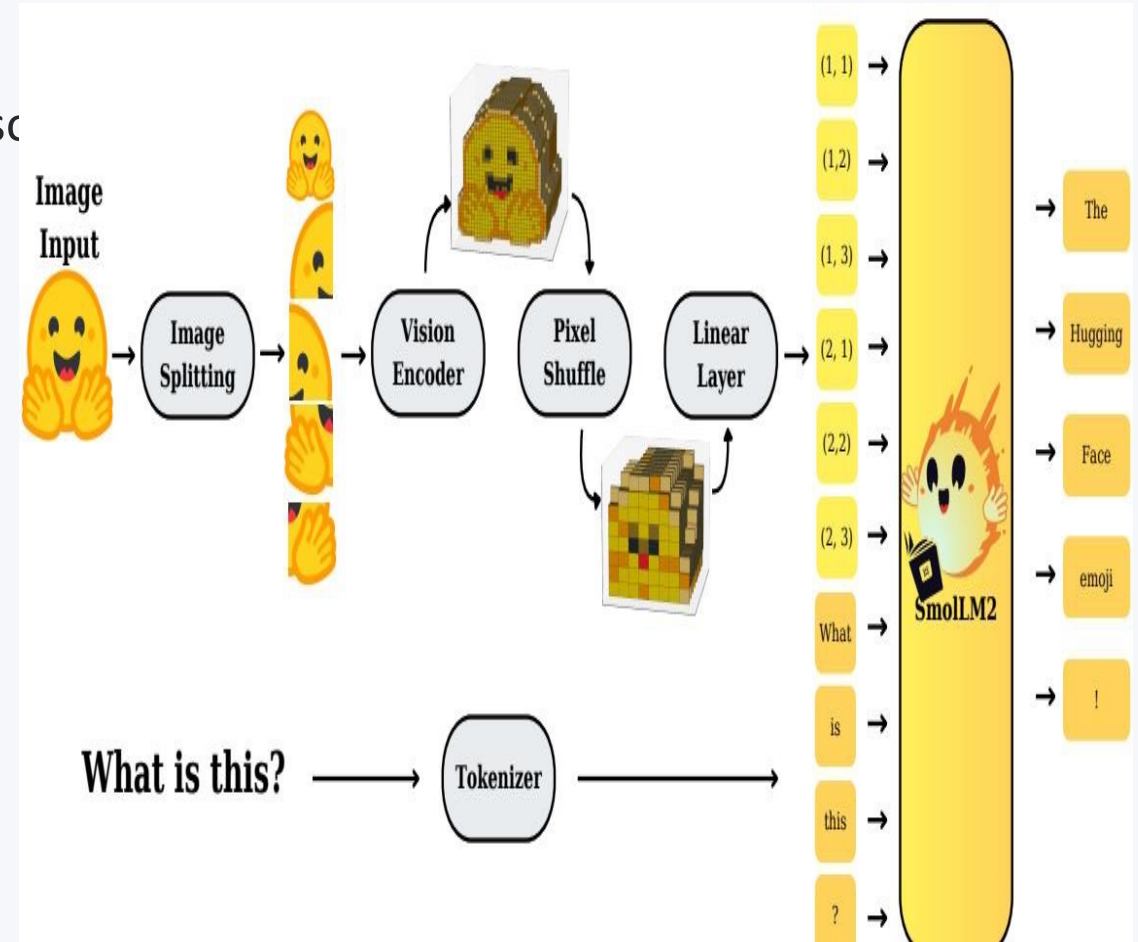


# SmolVLM: Redefining Small and Efficient Multimodal Models

Hugging Face × Stanford University

Compact VLMs enabling competitive multimodal reasoning



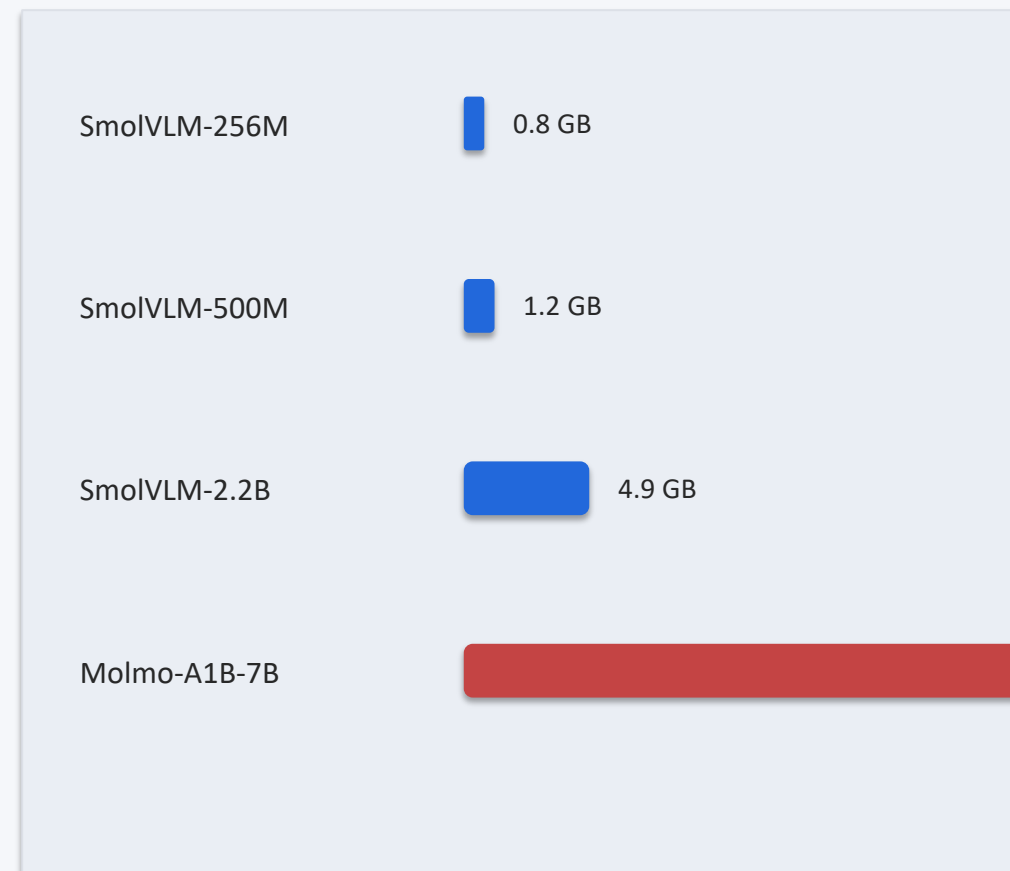
# Large VLMs Are Powerful but Impractical for Edge Deployment

State-of-the-art VLM performance often comes with prohibitive memory and compute costs.

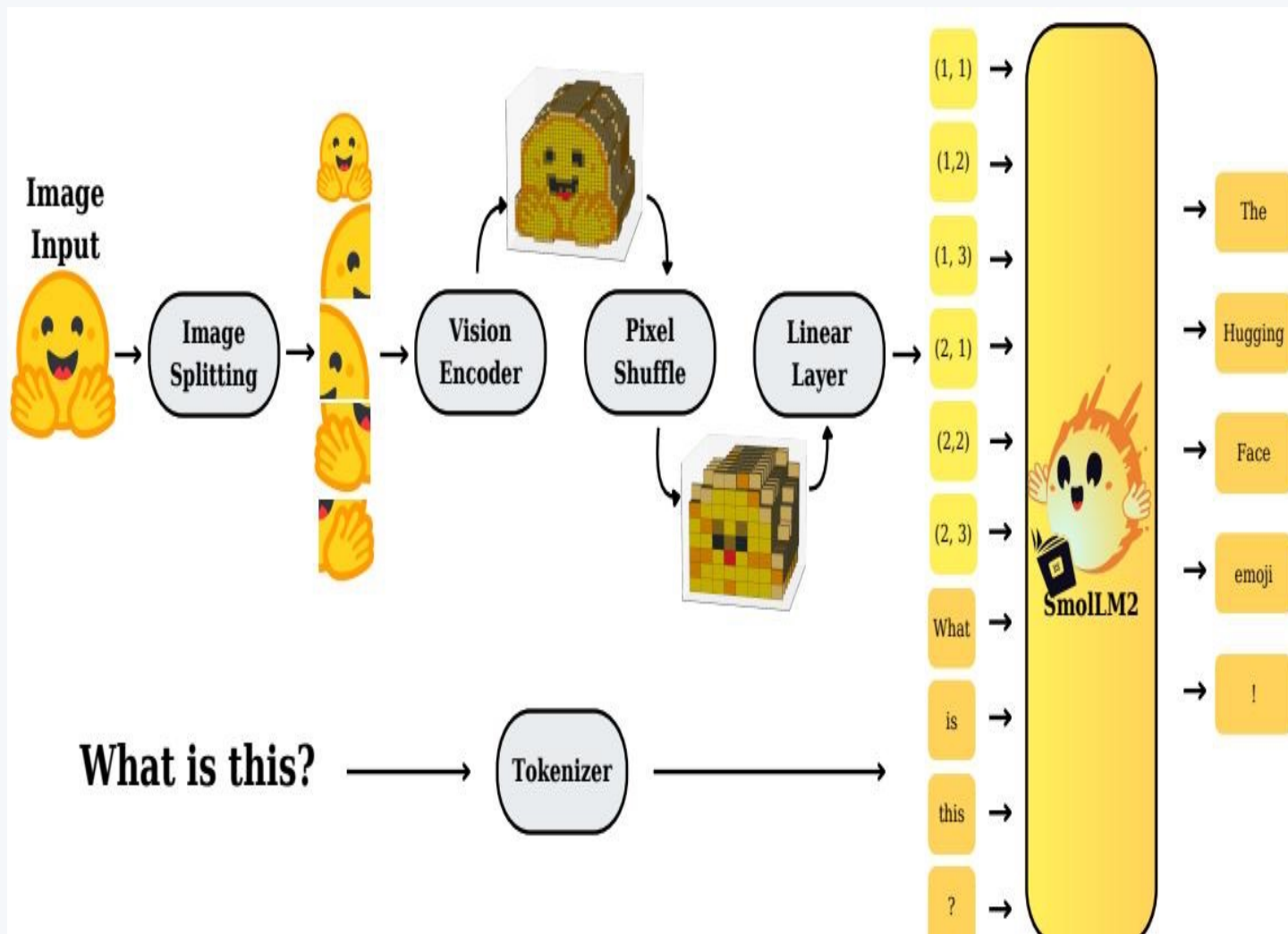
Edge and mobile deployments require tight RAM budgets and low-latency local inference.

SmolVLM targets this gap with 256M / 500M / 2.2B variants optimized for efficiency.

SmolVLM-256M runs at 0.8 GB RAM (batch=1), while Molmo-A1B-7B requires 27.7 GB.



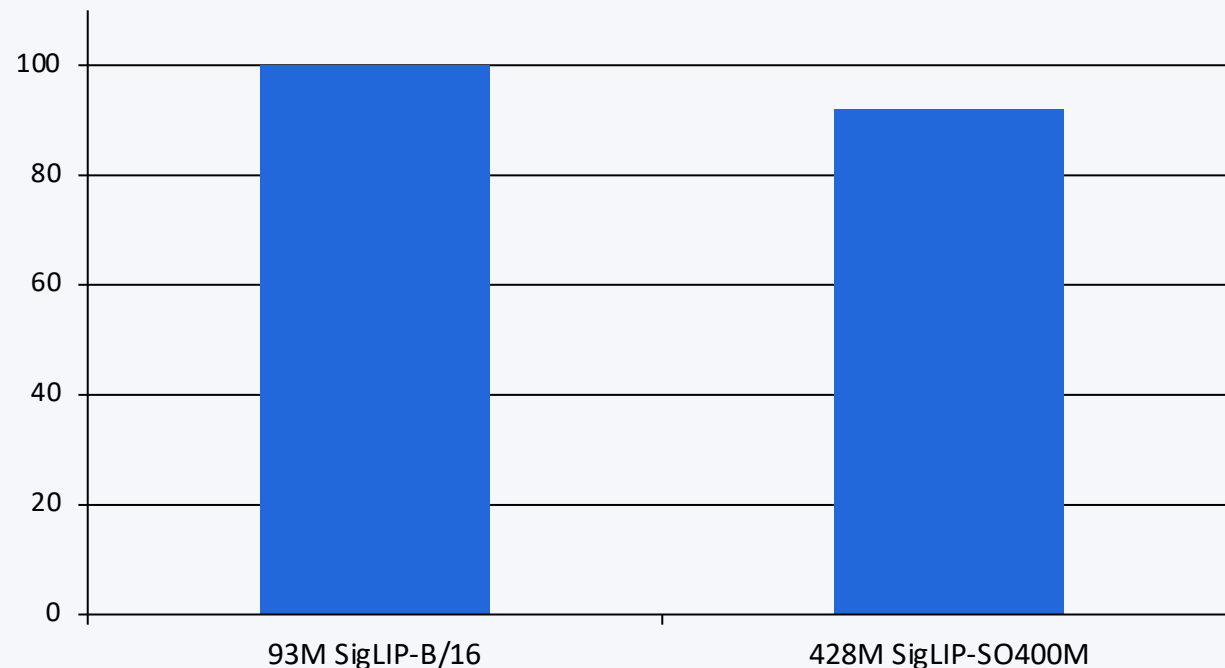
# SmolVLM Architecture: Image Splitting, Pixel Shuffle, and SmolLM



- 1) Split image(s) into sub-images for scalable visual p
- 2) Encode patches with SigLIP vision encoder.
- 3) Apply pixel shuffle (space-to-depth) for token com
- 4) Project to SmolLM2 embedding space and concate

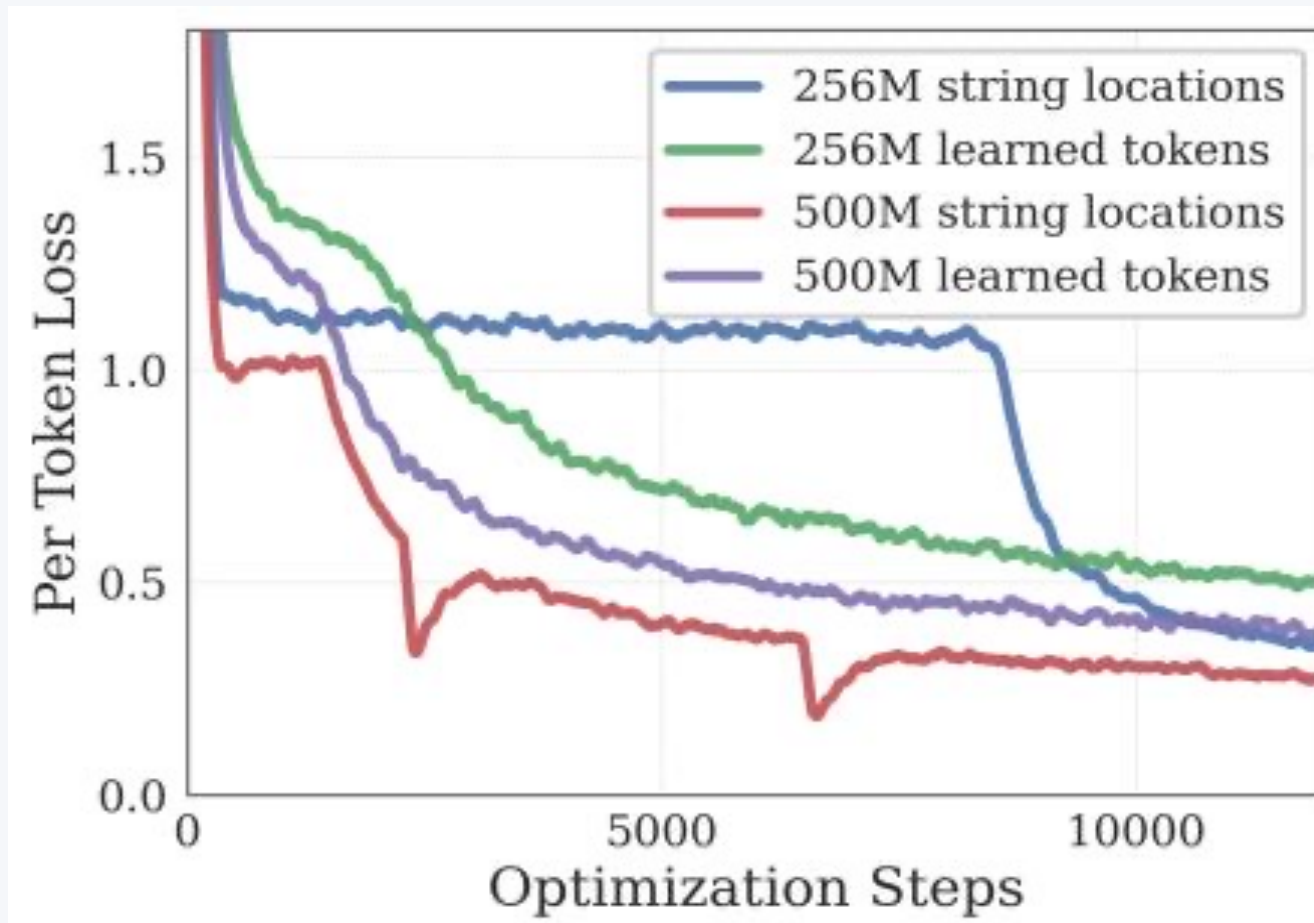
# Smaller Vision Encoders Outperform Larger Ones in Compact VLM

**Relative Result (small LM setting)**



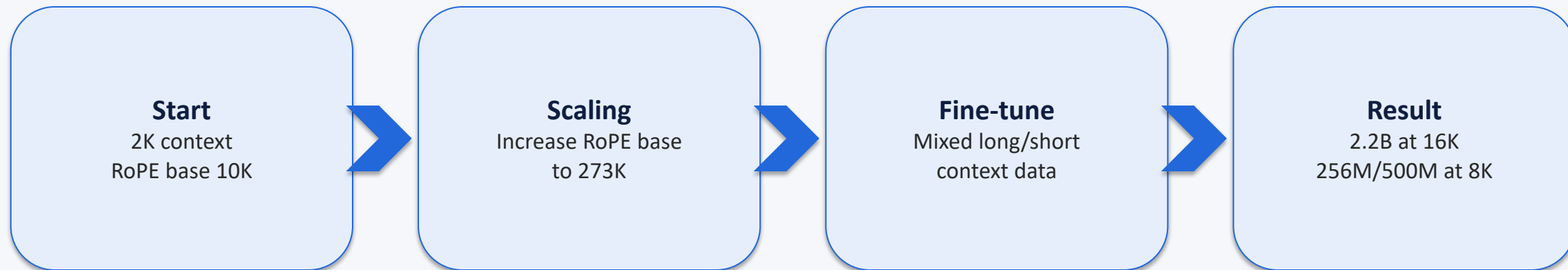
Finding 1: parameter allocation between vision encoder and L  
With the same 135M LM, 93M SigLIP-B/16 outperforms 428M  
Larger encoder only becomes worthwhile near 1.7B LM scale.  
Design implication: spend parameters where they maximize e

# Aggressive Pixel Shuffle (r=4) Cuts Visual Tokens 16× for Small Models



Pixel shuffle rearranges spatial dimensions into channels, reducing the number of visual tokens. At  $r=4$ , visual tokens are reduced by 16× versus no shuffle. Small models benefit from aggressive compression ( $r=4$ ) due to reduced memory usage. Larger models tend to prefer  $r=2$  for better fidelity/efficiency.

# Extending Context from 2K to 16K Tokens Unlocks Higher Image F

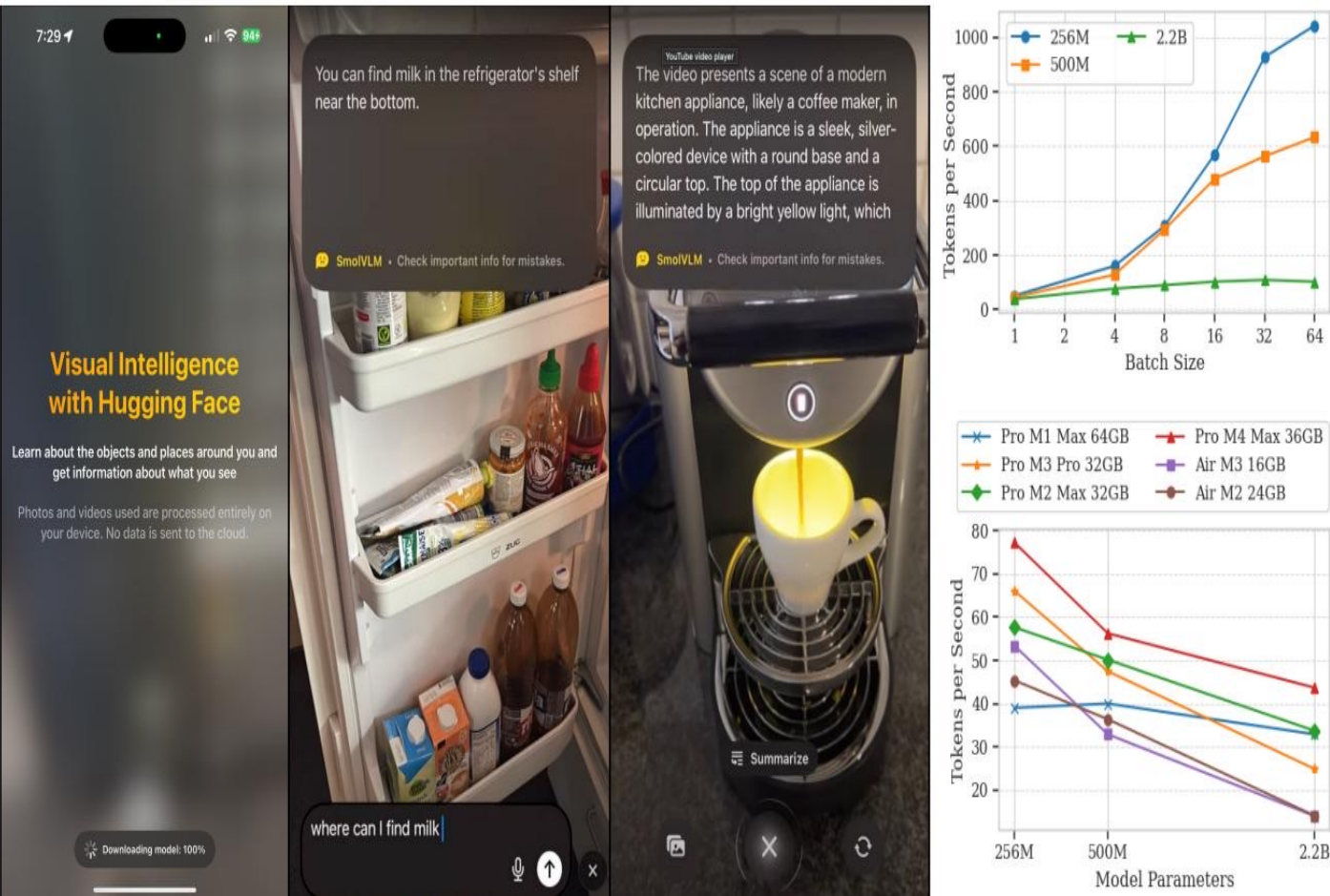


# SmolVLM Achieves Competitive Performance at Sub-1GB to 4.9G

| Model        | Average Score | RAM (GB) | Efficient OS comparator     | Comparator RAM (GB) |
|--------------|---------------|----------|-----------------------------|---------------------|
| SmolVLM-256M | 44.0%         | 0.8      | Molmo-A1B-7B / InternVL2-2B | 27.7 / n/a          |
| SmolVLM-500M | 51.0%         | 1.2      | Molmo-A1B-7B / InternVL2-2B | 27.7 / n/a          |
| SmolVLM-2.2B | 59.8%         | 4.9      | Molmo-A1B-7B / InternVL2-2B | 27.7 / n/a          |
| Molmo-A1B-7B | —             | 27.7     | Baseline efficient OS       | 27.7                |
| InternVL2-2B | —             | n/a      | Baseline efficient OS       | n/a                 |

Across 13 benchmarks, SmolVLM scales from 44.0% → 59.8% while staying within 0.8–4.9 GB RAM.  
SmolVLM-256M surpasses the 300× larger Ldefics-80B per reported key findings.  
Best Efficient OS references in paper: Molmo-A1B-7B and InternVL2-2B.

# Real-Time Inference on iPhones: Up to 80 Tokens/Second on App



HuggingSnap demonstrates fully local image/video understanding. SmolVLM-256M reaches ~80 tokens/s on Pro M1 Max 64GB. At batch=64, 256M approaches ~1000 tokens/s on high-end MacBook Air M3 16GB reports ~55 tokens/s for 256M model.

# Key Takeaways: Design Principles for Efficient Multimodal Model

Balance encoder and LM compute: smaller vision encoders can outperform larger ones in compact regimes.

Extend context length (8K/16K) with RoPE scaling to unlock better high-resolution multimodal understanding.

Use aggressive pixel shuffle compression ( $r=4$ ) for small models to reduce attention costs.

Sub-1GB multimodal inference is achievable without catastrophic performance loss.

Open models, data, and deployment tooling accelerate reproducible on-device AI research.

**Implication: architectural efficiency, not just scale, is the key lever for practical edge multimodal systems.**